MILITARY SPECIFICATION

PROJECTILE 155MM: XM864
PROPELLANT GRAIN

This specificaton is approved for use by the U.S. Army Armament Research and Development Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and preparation for delivery criteria for propellant grain used in XM864 155MM Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

MILITARY

MIL-STD-286	Propellants, Solid Sampling, Examination and Testing
MIL-STD-410	Non-destructive Testing Personnel Qualification and Certification

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

9381125	Propellant Grain

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

CHEMICAL PROPULSION INFORMATION AGENCY

CPIA Publication No. 21	JANNAF Solid Propellant Mechanical Behavior Manual (Section 4.3.2, Uniaxial Tensile Test of Constant Strain Rate; Section 4.7.5 Propellant Liner Peel Test, Nov. 70)

(Copies of the above documents required by contractors in connection with specific acquisiton functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 General. Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications. Materials shall be checked for shelf life in accordance with individual material specifications and government approved process controls (see 3.2).

3.2 Process controls. The propellant grain and inhibitor shall be fabricated in accordance with approved process controls. Following initial approval, any change in the approved process controls shall require prior Government approval (see 6.4).

3.3 Heat of reaction. The heat of reaction of each batch of propellant shall be 930±35 calories per gram.

3.4 Strand burning rate. When tested at 70 ± 5°F the strand burning rate will be 0.175 ± .009 inches per second at 500 psig and 0.210 ± .010 inches per second at 1000 psig.

3.5 Inhibitor peel strength. The minimum peel strength of the inhibitor/propellant bond at 70°±5°F shall be 8 pounds per/linear inch (p.l.i.).

3.6 Physical defects.

3.6.1 Inhibitor separation. On finished propellant segments there shall be no visual evidence of inhibitor/propellant separation (delamination) within one inch of the edge of any inhibitor/propellant mating surface. In addition there shall be no visual evidence of bubbles or separations in the inhibitor/propellant bond that are, individually, greater than .01 square inch in area or collectively greater than .02 square inch in total combined area.

3.6.2 Cast inhibitor defects. There shall be no visual evidence of perforation on the cast inhibitor.

3.6.3 Internal defects. The propellant segment shall contain no cracks, voids, or fissures measuring more than 0.030 inch in any direction. The total combined length of all cracks, voids, or fissures shall not exceed 0.060 inch and the combined length of all foreign material (maximum dimension each particles) shall not exceed 0.040 inch.

3.7 Mechanical properties. The cured propellant shall have the following properties at 70 ± 5°F:

a. Modulus of elasticity 1075 ± 375 psi.
b. Maximum stress of 170 psi minimum per sample average.
c. Strain at maximum stress 30% minimum per sample average
d. Ultimate strain 30% minimum per sample average

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3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.9 Proving ground tests. The grain shall comply with the following requirements when tested in accordance with 4.4.3.1.

3.9.1 Functioning. When assembled to projectiles the grain shall ignite, burn and enable the projectile to achieve its target range.

3.9.1.1 Horizontal range. No projectile shall have a horizontal range less than 97.5 percent of the mean horizontal range. Additionally, the range probable error (RPE) of the ballistic samples shall not exceed .441 percent when fired as specified in 4.5.6.

3.9.2 Security. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight.

3.10 Workmanship. Prior to casting, the batch of propellant shall be homogeneous and there shall be no streaks or discoloration signifying improper or incomplete blending of ingredients. The finished propellant grain shall be free of cracks, dirt, grease, gouges, foreign matter or any other physical defects which would render the item unsuitable for its intended use. Marking shall be complete and legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

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a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. When the contract requires first article inspection (see 3.8), the contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Description	Quantity Required
Spacers	20
Strand burning specimens	12
Bulk sample of sufficient size for 5 tensile and 2 heat of reaction specimen	1
Inhibitor peel strength specimens	2
Propellant grain (finished)	16

The first article sample shall be manufactured using the same approved process controls (see 3.2), equipment and procedures as will be used in subsequent production and all materials shall be obtained from the same source of supply as will be used in regular production. Test equipment and standards, as required, shall be approved by the acquiring activity prior to submission of the first article sample.

4.3.2 Inspection to be performed. See MIL-A-48078 and the first article inspection table herein.

4.3.2.1 Ballistic tests

4.3.2.1.1 Submission. Twelve (12) complete propellant grains packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government Installation as designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the propellant grains shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1).

4.3.2.1.2 Tests to be performed. Proving ground tests shall be performed using the procedures of 4.5.6.

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TABLE 1 FIRST ARITICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70505 (AR)

PARAGRAPH	TITLE		SHEET 1 of 1		DRAWING NUMBER
	Propellant Grain Components				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spacer (Dwg 9381126)				
	Examination for defects	20	-	3.1	4.4.2.1
	Propellant after mixing and curing (Dwg 9381125)				
	Mechanical properties	5		3.7	4.5.2
	Strand burning	12*		3.4	4.5.1
	Inhibitor peel strength	2*		3.5	4.5.4
	Heat of reaction	2*		3.3	4.5.3
	Grain split after curing, before Assembly (Dwg 9381125)				
	Examination for defects	12		3.1	4.4.2.3
	Grain after assembly (Dwg 9381125)				
	Radiographic inspection	16	-	3.6.3	4.5.5
	Ballistic test	12	-	3.9	4.4.3.1/4.5.6
	Examination for defects	16	-	3.1	4.4.2.4

NOTES:

*For First Article Inspection, this sample size supercedes quantity in the test method paragraph

DESC-QC (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.3.3 Rejection See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of grains shall have raw materials each from one lot of one manufacturer except as follows:

a. Zirconium carbide (ZrC) from no more than two lots from one manufacturer.

b. Dioctyl Azelate (DOZ) from no more than two lots from one manufacturer.

4.4.1.1 Propellant or inhibitor batch. A batch of material, either inhibitor or propellent, is defined as a homogeneous quantity of raw materials that are blended and processed in at one time. As a maximum a batch shall consist of materials blended at one time in one mixing or blending bowl. A batch shall contain individual ingredients from not more than one lot from one manufacturer for each ingredient.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105 as applicable.

b. Automatic gaging. Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with contract.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70505 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Spacer, Grain				9381126
					NEXT HIGHER ASSEMBLY 9381125
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		.40%	3.1	Gage
102	Diameter		.40%	3.1	Gage
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70505 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381125
4.4.2.2	Propellant after mixing and curing				NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mechanical properties	*		3.7	4.5.2
102	Strand Burning	*		3.4	4.5.1
103	Inhibitor Peel strength	*		3.5	4.5.4
104	Heat of reaction	*		3.3	4.5.3
Minor	None defined				

NOTES: *Sample size as specified in the applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70505 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9381125
4.4.2.3	Grain split after curing, before assy.				NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Special a	Inhibiter separation		100%	3.6.3	4.5.5
Major	None defined				
Minor 201	Evidence of poor workmanship		.65%	3.10	Visual

NOTES:

DRSMC (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 7[illegible] which may be used until exhausted.

AMENDMENT 2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70505 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Grain after assembly	1 of 1	9381125
			NEXT HIGHER ASSEMBLY: None

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Radiographic inspection *		100%	3.6.3	4.5.5
b	Inhibitor perforated, split or separated		100%	3.6.1	Visual
Major					
101	Inside diameter		.40%	3.1	Gage
102	Outside diameter		.40%	3.1	Gage
103	Corner radius		.40%	3.1	Gage
104	Total height across four segments		.40%	3.1	Gage
105	True position of inside diameter		.40%	3.1	Gage
106	True position of slots (4 places)		.40%	3.1	Gage
107	Spacer missing (8 places)		.40%	3.1	Visual
Minor					
201	Location of spacers		.65%	3.1	Gage
202	Evidence of poor workmanship		.65%	3.10	Visual

NOTES: * At the option of the contractor the grain may be radiographically inspected as individual splits in lieu of as an assembly.

DESC-FOA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70505 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Packaging, Grain				See 5.0
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag damaged		.40%	5.1	Visual
102	Drum split or damaged		.40%	5.1	Visual/Manual
103	Tape missing or improperly applied		.40%	5.1	Visual
Minor					
201	Marking incomplete or incorrect		.65%	5.1	Visual
202	Evidence of poor workmanship		100%	3.10	Visual

NOTES:

DD FORM 1664-R-1 (D) Form 160, 1 Aug 83 replaces edition of 1 Jul ~~~ which may be used until exhausted.

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4.4.3 Testing. Testing is described in the Quality Conformance and First Article Inspection tables and herein.

4.4.3.1 Proving ground. These tests shall be performed as specified in 4.5.6. Defects shall be classified as follows:

Critical.

1. Grain failure inbore.

Special.

a. Range less than 97.5 percent of mean horizontal range.
b. Burner assembly fails to ignite.

Major.

101 Range probable error exceeds allowable limit.

Minor: None defined.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.5 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as requried by the above paragraph.

4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5 Test methods and procedures.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Strand burning characteristics of propellant. Along with propellant grains cast from each batch or mix of chemicals, six test strands shall be taken from the same propellant batch, cured in the same curing chamber under the same conditions for the same

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length of time and subjected to this test in accordance with Method T803.1 of MIL-STD-286. Three strands will be tested at 500 psig and three at 1000 psig. If any specimen fails to meet the requirements of para 3.4, the batch shall be rejected.

4.5.2 Mechanical properties of propellant. Along with propellant grain cast from each batch or mix of propellant, one physical property sample sufficently sized shall be cast from the same propellant batch, cured in the same curing chamber under the same conditions for the same length of time and subjected to these tests in accordance with the JANNAF Class B procedures as described in CPIA Publication #21 "Uniaxial Tensile Tests" at constant crosshead rate. Test specimens shall be conditioned at the applicable test temperature for a minimum of two hours prior to performing the mechanical property tests. The crosshead speed of the tensile tester shall be 2.0 inches per minute and the chart speed shall be 10 inches per minute. Test specimens shall be cut from a bulk sample. Five (5) test specimens shall be prepared and tested. If the average value of the five test specimens fail to meet the requirements of 3.7, the batch of propellant of which it is representative shall be rejected.

4.5.3 Heat of reaction. One representative propellant specimen obtained from the same physical property sample prepared for the mechanical property tests shall be used to perform this test in accordance with Method 802.1 of MIL-STD-286 or an approved equivalent. The 3 gram sample and oxygen purge requirements shall not apply.

4.5.4 Inhibitor peel strength. Initially each batch of inhibitor will be tested with each batch of propellant that it is utilized with. A representative sample of each batch of inhibitor material or part thereof shall be bonded to a representative sample from the propellant batch to which it is bonded in a finished grain. This sample shall be cured in the same curing chamber under the same conditions for the same length of time and be tested in accordance with section 4.7.5 of CPIA Pub. 21. Two test specimens shall be tested for each batch of propellant. If any specimen fails to meet the specified requirement the individual propellant/inhibitor combination material batch shall be rejected.

4.5.4.1 Reduced Sampling. After successful acceptance of ten inhibitor material batches with each propellant batch to which it has been bonded, the inhibitor peel strength sampling shall be reduced to one propellant/inhibitor combination batch per inhibitor material batch. Failure of any specimen shall require rejection of the inhibitor batch and all grains bonded to that inhibitor batch. Sampling will revert to the initial sampling sequence until reduced sampling can again comply with the above reduction criteria.

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4.5.5 Physical defects. The finished propellant grain shall be radiographically inspected for physical defects in accordance with approved procedures, first, with the beam parallel with the longitudinal axis; second, with the beam directed radially from the perpendicular to the central longitudinal axis of the grain when assembled. A penetrometer or sample test block showing the types of defects for rejection shall be designed by the contractor to assure a two percent minimum sensitivity and approved by the acquiring activity (see 6.5). Finished propellant grain assemblies which exhibit any of the physical defects specified in 3.6.2 shall be rejected and removed from the lot.

4.5.6 Proving ground. The grains sampled for this test shall first be loaded into live base burner assemblies; then be forwarded to a load, assemble and pack facility for assembly onto live loaded projectiles; and forwarded to a designated government proving ground for firing. This test shall be conducted in accordance with an applicable proving ground test procedure. All projectiles fired for this test will be assembled with a M557 point detonating fuze and a live self registration charge.

The projectiles fired in this test shall be fired in the M199 Cannon using a M203 series propelling charge. The projectile, fuze and propelling charge shall all be at ambient temperature. A constant gun QE will be maintained throughout the test. Pressure time traces will be made for each projectile fired. Smear photographs will be taken at 10 and 30 meters in front of the muzzle. Doppler radar (HAWK) will be used.

An effort will be made to fire all projectiles for this test within a one hour time span. Observation shall be made to determine compliance with 3.2. Stop the test if surface winds exceed 20 knots or if two consecutive wind speed readings differ by more than 15 knots or if the wind direction of two consecutive readings differ by more than 800 mils (does not apply if wind speed is less than 5 knots).

4.5.6.1 Weapon and component data. All weapon and ammunition components used in support of this test shall be identified by model and lot or serial numbers.

4.5.6.2 Data reporting. As a minimum the proving ground test report will include the following observable attribute data:

a. Premature functioning
b. Metal parts separation
c. Impact functioning (information only)
d. Round by round range.

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4.5.10.3 Calculation of percent RPE.

$$\text{Percent RPE} = \frac{67.45}{X}\sqrt{\frac{\sum_{i=1}^{10}(X - X_i)^2}{9}}$$

Where: x = Average range of group tested
x_i = Individual projectile range

4.5.10.4 Sample size. Twelve grains (assembled) will be randomly selected for this test. Ten grains (assembled) are to be fired and counted. The remaining two grains (assembled) are available for use of the proving ground should any of projectiles fired be considered invalid due to test conditions. (Note: The required RPE assumes a ten round sample.)

5. PACKING

5.1 Packaging, packing and marking. Packaging, packing and marking of grains shall be in accordance with either 9386975 (level B) or 9386976 (level A qualified) in accordance with the contract. (This section is not applicable if Base Burner LAP operations are to be performed at the same facility).

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use in Base-Burner Assemblies for XM864 155MM Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. First article inspection, when required (see 3.8).

6.2.2 Contract data requirements Contractor engineering documentation specified below will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

Paragraph	Data Requirement	Applicable DID
3.2	Process control	DI-P-1604
4.4.4	Inspection equipment approval	DI-R-10054

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(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Objective evidence.

6.3.1 General. Inspection records and documentation verifying compliance with applicable drawing and specification requirements.

6.3.2 Materials. A statement (certification), supported by test data, that materials produced or purchased by the contractor meets all requirements when such material is controlled by Government or commercial specifications referenced in contractual documents.

6.4 Process controls. Process controls and process control changes should be submitted for approval to Commander, US Army Armament Research and Development Center, ATTN: SMCAR-LCU-SI, Dover, NJ 07801-5001. Process controls shall specify in detail all materials, procedures, techniques and equipment (including handling of ingredients, mix cycle, sequences of addition of materials, mixing times and conditions, cure temperatures and times etc.) required to process, fabricate, inspect and test a quality product suitable for its intended use as determined by the acquiring agency. Detailed test and inspection record format shall be included as an integral part of the process and shall be in sufficient detail as to enable easy verification by the Government QAR designated by the acquiring agency.

6.5 Equipment design approval. Submit equipment designs, as required, to Commander, US Army AMCCOM, ATTN: AMSMC-QAR-I(D), Dover, NJ 07801-5001.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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